

Javier Martinez

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EDUCATION

Georgia Institute of Technology (GT), Atlanta, GA

June 2024 – May 2028 (*Expected*)

B.S. Mechanical Engineering

WORK EXPERIENCE

GTRI Agriculture Technology Research Program, *Mechanical Engineer Student Assistant*, Atlanta, GA February 2026 – Present

- Designed and fabricated an automated tile cleaning table for chicken feces collection for tile that are 15" x 24" x 3/8", cutting per tile cleaning time from 15 to 20 minutes by hand to under 30 seconds, reducing 8 hours of labor into 1 hour of labor
- Engineered a 10' x 7' poultry-coop floor system, with a 5 x 6 tiles arrangement that collects feces of chickens. Tiles were labeled to map out high feces location in the coop flooring system. An automated tile pusher system developed using 3 linear actuators to push 2 tiles out of the poultry-coop flooring system was designed to collect and record feces data
- Converted a 4-stepper motor small scale model from an unreliable open-loop system to a closed-loop system by replacing driver boards to use built-in encoders and created an electrical box, improved positioning accuracy and space optimization for electronics and easy future trouble shooting

RESRG (Automotive Manufacturing), *Industrial Engineering Intern*, Covington, GA

May 2025 – August 2025

- Ran detailed time studies on two robotic press cells; mapped operator tasks, robot cycles, and changeovers to improve availability performance. Increased one cells production by 15% and corrected an underperforming cell running 18% less than its standard
- Built an Excel-based OEE tracker (availability, performance, quality) of 20-25 cells and records of current time cycles to find bottlenecks within these cells and serves as a tracking tool
- Designed and assembled a 9' x 2' x 6' flow rack system for one cell to eliminate the travel distance operators must travel to switch filled and empty tote boxes. This allowed operators to stay in one area and decreased their cycle time by 10 seconds and improved their OEE from 70% to 81%

PROJECTS

Design & Manufacture Weaponized Robots, *RoboJackets BattleBots*, Atlanta, GA

August 2024 – Present

- Designed a vertical beater bar, for 3lb class, out of 4130 steels; manufactured with water jet, milling. Redesigned after machining errors by switching with aluminum 6061 as a last resort before competition
- Competed in 2 official matches; drivetrain failures led to losses, but from this a future redesign is planned for next season
- Designed a 12-13lb weapon-subsystem. Consisting of the "Drumlet" weapon, steel tube axel, aluminum pulley, and steel plates. Currently in the process of starting manufacturing all of the weapon-system components.

Creative Design & Decision, *Class Project*, Atlanta, GA

May 2026 – Aug 2026

- Created a dual-motor, one-way-bearing drivetrain and cascade lift in SolidWorks, converting 100 RPM input into 200 RPM drive and 33 RPM lift, reaching 101" height on a \$120 budget
- Led a 4-person team as project manager and fabricator, building the system with laser cutting and 3D printing, placing 3rd in class competition

LEADERSHIP

Team Lead "Quantiti over Qualiti", *RoboJackets BattleBots*, Atlanta, GA

May 2026 – Present

- Leading a 15-person team for 30lbs weight class division, with the design goal of creating "multi-bots", 2-20lb robots
- Held 4 meetings a week for the summer, delegated design task, deadlines before starting fabrication

Sub-Team Lead Weapon System, *RoboJackets BattleBots*, Atlanta GA

June 2025 – May 2026

- Led a 4-person sub-team for the weapon subsystem, running weekly meetings, assigning design tasks, and reporting at sub-system meetings
- Designed a "Drumlet" weapon that weighs 11-12lb of steel 4140 and spans across 13 inches, for the 30lb robot.

SKILLS

Design: Inventor, SolidWorks, Fusion 360, Revit, OnShape,

Fabrication: 3D Printing, Lathe, Milling, Waterjet, CNC Mill, Bandsaw, Mechanical Assembly,

Programming: MATLAB, Arduino, C++

Languages: English (native) Spanish(fluent)

Additional: AutoCAD, Inventor, Revit, and Fusion Certified